Introduction to NYT [1]

Dirk Riehle, FAU Erlangen

NYT A01

[1] Nailing Your Thesis

Course Learning Goals

General objectives (NYT-VUE)

- Understand science in general
- Understand the scientific process, including
  - Research process and designs
  - Research methodologies and methods
  - Publications and the scientific community
- Learn how to write a research paper (thesis)

Project objectives (NYT-PROJ)

- Learn by performing a small research project

Definition of Science (Working Definition)

Science is the **process of acquiring knowledge** for **correct prediction and reliable outcome.** [DR]

Course Content and Structure

Overview

1. What is science?
2. Scientific research

Theory building

3. Qualitative data analysis
4. Systematic reviews
5. Qualitative surveys
6. Action research
7. Case study research

Theory validation

8. Survey research
9. Controlled experiments

Comprehensive

10. Design science

Academia

11. Academic writing
12. Academic publishing

Skills Required for Course

No particular requirements, but …

- Strong conceptual and analytical thinking

Requirements for Final Theses

From the Bachelor-Prüfungsordnung Informatik (“writing about science”)

- “Die schriftliche Bachelorarbeit soll ein wissenschaftliches Thema aus dem Bereich der Informatik behandeln.”

From the Master-Prüfungsordnung Informatik (“applying results of science”)

- “Die Masterarbeit dient dazu, die selbständige Bearbeitung von wissenschaftlichen Aufgabenstellungen der Informatik nachzuweisen.”

From the Promotionsordnung Informatik (“creating scientific progress”)

- “Die Dissertation soll die Fähigkeit des Bewerbers belegen, ingenieurwissenschaftliche Probleme selbständig und mit Erfolg zu bearbeiten und Wege zu ihrer Lösung zu finden.

Course Position in Curriculum

Bachelor's						Master's			
1	2	3	4	5	6	1	2	3	4
				ADAP	NYT		AMOS	AMOS	
				FOSS	FOSS		COACH	COACH	
				SEMI	SEMI	SEMI	SEMI	SEMI	MATH
				PRAK	PRAK	PRAK	PRAK	PRAK	
					BATH				
					KOLL				KOLL

Courses and Modules

		Courses (Lehrveranstaltungen)			
		Lecture (2 SWS)	Exercise (2 SWS)	Project (2 SWS)	Total ECTS
Modules	NYT-VUE	x	x		5
	NYT-PROJ			x	5
	NYT-VUE+PROJ	x	x	x	10

VUE = Lecture + exercise (Vorlesung + Übung)
PROJ = Project

Availability of Modules

		University			
		FAU Erlangen			
Modules	NYT-VUE	x			
	NYT-PROJ	x			
	NYT-VUE+PROJ	x			

x = available
– = not available

NYT-VUE Grading [1]

Method exercises = 100% (of semester contributions)

- Miss more than one exercise and you fail the class

[1] See https://oss.cs.fau.de/teaching/course-resources/grading-schemes-and-scales/

Method Exercises (Qualitative Data Analysis)

Graded using a structured metric (intercoder reliability) against our solution

One qualitative written assignment (presentation of resulting theory)

Grading Rules

All homework is individual work

No Oral or Written Exam [1] [2]

[1] If both you and we don't want to
[2] You still have to register for the course

Course Language [1]

Class

- Lecturer: English
- Student: Choice of German or English

Exercise

- Instructor: English
- Homework: Choice of German or English

Project

- English only

[1] See https://oss.cs.fau.de/2012/03/10/english-or-german-deutsch-oder-englisch/

The Three Steps to Proper Registration

1. **Course registration**
   a. Sign-up on [StudOn](#) / open now
   b. Needed to get your id for matching you on Campo
2. **Course management**
   a. Sign-up on [Moodle](#) / open now
   b. Use your university email address; if not no grade
3. **Exam registration**
   a. Sign-up on [Campo](#) / opens later
   b. No exam registration no grade

Course Participation (Auditing)

The course is provided publicly, you can always audit it

Auditing will not give you credit points towards a degree

Work Rhythm

Lecture session (class)

- Review of last week (quiz)
- Presentation of this week’s topic

In-class method exercise

- Discussion of articles
- Discussion of homework

Self-study / homework

- See Course organization doc
- Self-organized

Course Communication

Announcements by email

- Through course management system

Questions to teaching team

- Please ask your question in the course forum
- For private questions, use the teaching team email alias

Non-urgent questions will be answered in class

Thank you! Any questions?

dirk.riehle@fau.de – https://oss.cs.fau.de

dirk@riehle.org – https://dirkriehle.com – @dirkriehle

Legal Notices

NYT-VUE Grading [1] (Old)

Lecture-time contributions = 50% of total grade

- Lecture content (4 SWS) = 60 / 150 = 40% (of semester contributions)
  - Graded using [0..10] each class session using class quizzes
  - Miss more than three quizzes and you fail the class
- Method exercises = 90 / 150 = 60% (of semester contributions)
  - Graded using [0|1|2|3] using tool and grading rubric
  - Miss more than one exercise and you fail the class

Oral exam = 50% of total grade

- If both you and we agree on not having an oral exam, it can be dropped
- The oral exam is scheduled during class on the last class day
  - You need to tell us at least two weeks in advance if you want an oral exam

[1] See https://oss.cs.fau.de/teaching/course-resources/grading-schemes-and-scales/

Class Quizzes (Old)

Each class session starts with a class quiz

- A quiz will test your understanding of last session's topic
- A quiz typically has 5 questions and will last 10 minutes
- The overall quiz is graded using [0..10] scheme (10 points in total)

A class quiz will open precisely when class starts

- The quiz is administered automatically
- It is your job to have reliable Internet access etc.
- There is no way to make up for a missed quiz

Grading Rubric for Homework Submissions

Categories	Points					Criteria
	Disagree	Disagree some	Neutral	Agree some	Agree	
Form (10%)	**Does the deliverable meet formal requirements?**					Does it meet length requirements, is it written in the right language, etc.?
Language (10%)	**Is the language clear, concise, and helpful?**					Are sentences complete, is the grammar correct, are statements coherent, etc.?
Structure (20-30%)	**Does the deliverable have a clear logical structure?**					Does it follow established or suggested structure? Is the argument logical?
Content (50-60%)	**Does the discussion touch on all relevant issues?**					As a reference, use your own deliverable as well as what you learned in class.